

See it. Report it. Improve your city.

A computer-vision platform that turns one citizen photo into a verified, rated and routed civic complaint for potholes, illegal parking and garbage dumping.

Project report and pitch brief · Final-year AI & ML project · St Joseph Engineering College (SJEC), Mangaluru · October 2026

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3 civic issues

Pothole · Illegal parking ·
Garbage dumping

6 vision models

Detection, segmentation,
open-vocabulary and OCR

3 departments

Roads · Traffic · Sanitation,
routed automatically

1 photo

From the street to an
officer's queue in about a
minute

One photo, one verified complaint

Citizens see civic problems every day but rarely report them, because reporting is slow, vague and goes nowhere. CityLens fixes the first mile and closes the loop.

A citizen takes or uploads a photo. CityLens reads the location from the photo, and its computer-vision pipeline works out **what** the problem is, **where** it is in the image and **how serious** it is. It then sends the complaint to the **right department**. The citizen confirms the result before anything is filed. The officer in charge sees a ranked queue with the photo, the outlined problem, the map position and an explanation of the severity. They update the status, and they upload an "after" photo as proof of the fix.

6

AI models working together: three detectors, a street-scene segmenter, an open-vocabulary detector and an OCR reader

3

kinds of computer vision: object detection, instance and semantic segmentation, and text recognition

82

automated tests covering detection logic, cross-checks, security and the complaint workflow

What makes it different

- **It understands the scene, not just objects.** A car isn't automatically "illegal parking". CityLens checks whether it is on the footpath, inside a no-parking zone, under a NO PARKING sign, or blocking the road.
- **Its severity can be explained.** Every rating comes with a reason in plain words. For example: *"Garbage covers about 24% of the photo across 6 piles. Measured from the segmentation mask."*
- **The models check each other.** A pothole that lies on top of a detected car, or off the road surface, is rejected as impossible.
- **A human always makes the final call.** The citizen confirms or corrects the AI, and the officer acts. Disagreements become data for improving the models.

02 · THE PROBLEM

Civic reporting is broken at the first mile

For citizens

- They don't know which department handles which problem.
- Phone and web forms ask for typed addresses and categories.
- Nothing visible happens after reporting, so people stop reporting.

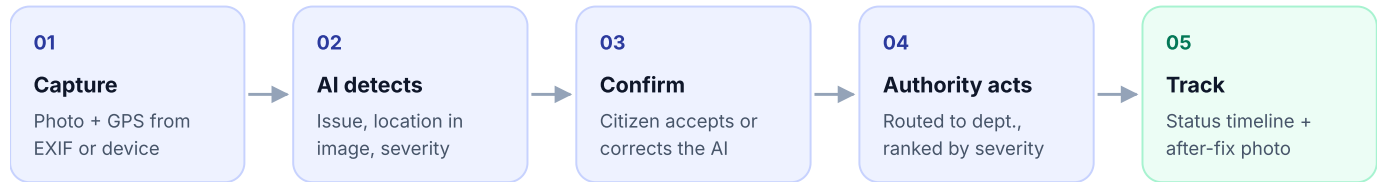
For authorities

- Complaints arrive vague and unverified, often duplicated, and in the wrong inbox.
- There is no objective way to rank the urgent ones first.
- Field staff spend time confirming reports instead of fixing them.

Existing government apps such as **Swachhata** already let people upload a geotagged photo of a sanitation problem. The photo is reviewed by people, though: the app does no automatic image analysis, and each app covers one department ([Vikaspedia](#)). CityLens adds the missing intelligence layer: automatic recognition, severity rating and routing across departments.

A five-step loop between citizen and city

Two web portals share one backend: the citizen app and the authority portal.



Citizen app

- Upload or capture a photo. GPS is read from the photo's EXIF data, or taken from the device.
- The address is filled in automatically (reverse geocoding with OpenStreetMap Nominatim).
- The AI result appears with outlines drawn on the photo, a confidence score, the severity and its reason.
- The citizen confirms the result, or picks the correct issue type.
- They get a complaint ID such as **CL-2026-00123** and can track its status as it changes.

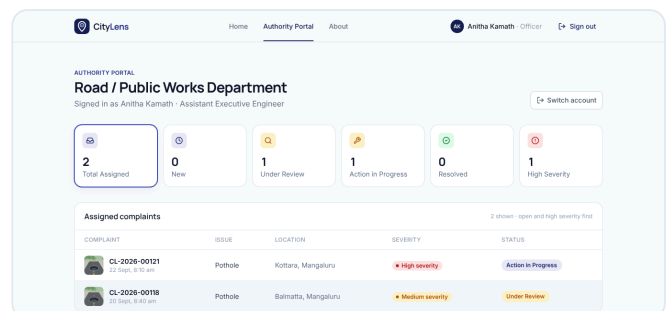
Authority portal

- Officers sign in to their own department and see only that department's queue.
- Open and high-severity complaints are listed first, with counts per status.
- A map shows every complaint at its GPS position.
- Each complaint moves through the stages *New* → *Under review* → *Action assigned* → *In progress* → *Resolved*, or is rejected with a reason.
- Officers add action notes and upload an "after" photo as evidence.

Automatic routing

ISSUE	DEPARTMENT	EXAMPLE OFFICER
Pothole	Road / Public Works Department	Assistant Executive Engineer
Illegal parking	Traffic / Municipal Enforcement Department	Traffic Sub-Inspector
Garbage dumping	Municipal Solid Waste / Sanitation Department	Health Inspector

The server decides the routing, not the client app, so a complaint can't be pushed into the wrong department's queue.



Authority portal: the Roads department's queue, sorted with open and high-severity complaints first.

The technology stack

The stack is open source end to end. Every model is pretrained and publicly available, so a city pays no licence fees.

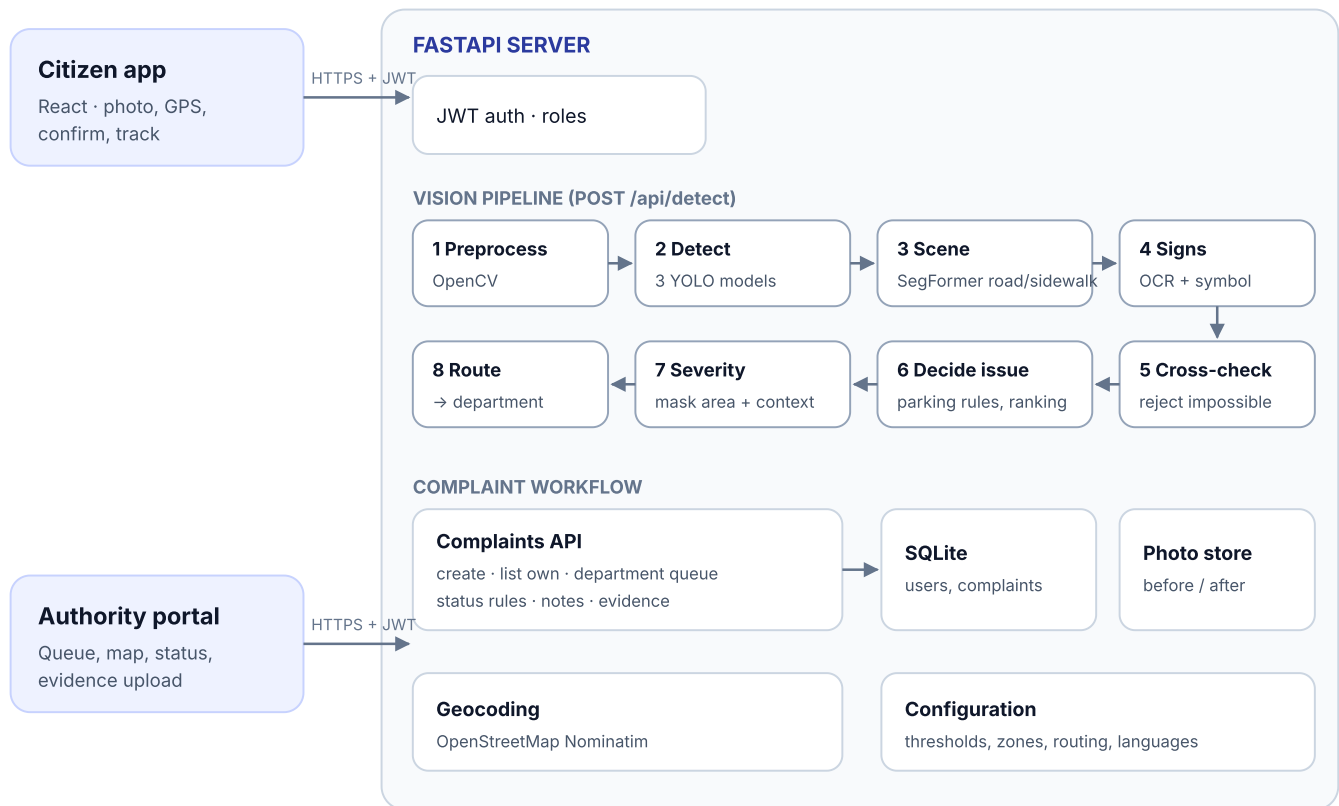
LAYER	TECHNOLOGY	WHAT IT DOES IN CITYLENS
Frontend	React 18, Vite, Tailwind CSS, React Router	Citizen app and authority portal; responsive on mobile and desktop, with a consistent animation system
	Leaflet + OpenStreetMap, exifr	Complaint map in the authority portal; reading the GPS position stored inside the photo
Backend	Python, FastAPI, Uvicorn	REST API for detection, complaints, the authority workflow and authentication
	SQLite	Users, complaints and status history (can be swapped for PostgreSQL in production)
	JWT (PyJWT, HS256) + PBKDF2-SHA256	Sign-in tokens and salted password hashing
Image processing	OpenCV, NumPy, Pillow	Decoding, EXIF rotation, resizing, contrast enhancement, colour segmentation, mask geometry
Object detection	Ultralytics YOLOv8 (PyTorch)	Finds potholes and vehicles: box, class and confidence
Instance segmentation	YOLOv8-seg	Pixel-accurate outline of each pothole or garbage pile, used to measure real area
Open-vocabulary detection	YOLOE + MobileCLIP text encoder	Garbage detector built from text descriptions, with no labelled training set needed
Semantic segmentation	SegFormer-B0 (Cityscapes), Hugging Face Transformers	Labels every pixel as road, sidewalk, building and so on, to judge where a vehicle stands
OCR	EasyOCR (English + Kannada)	Reads NO PARKING boards in the photo
Geocoding	OpenStreetMap Nominatim	Turns GPS coordinates into a readable address

The six models in the pipeline

MODEL	TYPE	TRAINED ON	SIZE	USED FOR
YOLOv8s-seg pothole	Instance segmentation	Public pothole dataset	~23 MB	Pothole outline and area
YOLOE-11s-seg (garbage prompts)	Open-vocabulary segmentation	Large-scale image-text data	28 MB	Overflowing garbage, garbage bags, litter
YOLOv8s	Object detection	COCO (80 classes)	22 MB	Cars, motorcycles, buses, trucks
SegFormer-B0	Semantic segmentation	Cityscapes street scenes	~15 MB	Road vs footpath, road-surface check
EasyOCR CRAFT + recogniser	Text detection + recognition	Multilingual text	~100 MB	Reading sign boards
Sign-symbol detector	Classical CV (colour + geometry)	No training needed	n/a	The no-parking symbol (blue disc, red ring and slash)

How the system fits together

The browser apps talk to one FastAPI server. That server runs the vision pipeline, stores complaints and enforces who may see and do what.



The request flow for one photo

1. The browser sends the photo and its GPS coordinates to **POST /api/detect** along with the user's sign-in token.
2. OpenCV decodes the image and fixes phone rotation. It then resizes the image to at most 1280 px, keeping the scale factor so results map back to the original pixels.
3. The three detectors run. If a vehicle is found, the scene segmenter and sign reader run too. They are skipped otherwise, which saves time.
4. The analysis step combines everything: cross-checks, parking rules, choosing the issue, severity and department.
5. The response contains the issue, confidence, severity and reason, each detection's box and mask outline, any signs read, and the department.
6. After the citizen confirms, **POST /api/complaints** stores the complaint with its owner (taken from the token), the photo and the AI evidence.

How the computer vision works

CityLens doesn't use a single model. It uses a pipeline of complementary vision techniques, each answering a different question about the photo.

QUESTION	TECHNIQUE	OUTPUT
What objects are in the photo, and where?	Object detection (YOLOv8)	Bounding boxes, classes, confidence
What exact shape and area does each problem have?	Instance segmentation (YOLOv8-seg, YOLOE)	A pixel mask per object
Which parts of the street are road, and which are footpath?	Semantic segmentation (SegFormer)	A class label for every pixel
Can we find garbage without a labelled dataset?	Open-vocabulary detection (YOLOE + CLIP)	Detections from text prompts
Does a sign forbid parking here?	OCR (EasyOCR) + colour/shape analysis (OpenCV)	Sign text and the sign's position
Do the answers make sense together?	Reasoning across the models (geometry)	Impossible detections rejected

Step 1 · Preprocessing with OpenCV

Phone photos arrive in every size and orientation. The image is decoded and rotated upright using its EXIF orientation tag. If its longest side is over 1280 px it is scaled down, which keeps detection fast without losing small potholes. Optional CLAHE (contrast-limited adaptive histogram equalisation) can brighten dark night-time photos. The scale factor is kept so every result is reported in the original photo's pixel coordinates.

Step 2 · Object detection with YOLO

YOLO ("You Only Look Once") is a single-stage convolutional neural network. It looks at the whole image once and predicts all objects in one pass, which is why it is fast enough for real-time use.

- A **backbone** (CSPDarknet) extracts visual features, such as edges, textures and shapes, at several scales.
- A **neck** (a feature pyramid network) merges fine detail with wider context, so both a small pothole and a large truck are found.
- A **detection head** predicts a box, a class and a confidence score at each location. YOLOv8 is *anchor-free*: it predicts box centres directly instead of adjusting preset box shapes.
- **Non-maximum suppression** removes duplicate boxes around the same object.

Vehicles are detected with **test-time augmentation**: the model also looks at flipped and rescaled copies of the photo and merges the results. On a top-down photo of ten parked cars, this raised the number found from 7 to all 10.

Step 3 · Instance segmentation: measuring real area

A bounding box overstates irregular shapes. A crescent-shaped pothole can fill well under half of its box. The segmentation versions of YOLO add a **mask head**. It predicts a small set of shared "prototype" masks for the image, plus coefficients for each object, and combines them into a pixel-accurate outline for every object.

- **Pothole area** is computed from the outline with the shoelace formula, so severity reflects the damage that is actually there.

- **Garbage area** is the *union* of all garbage masks, drawn onto a canvas with `cv2.fillPoly`, so overlapping piles are counted only once.
- The outlines are sent to the app and drawn on the photo. The citizen sees exactly what the AI found.

Step 4 · Open-vocabulary detection for garbage

There is no good public dataset of *dumped or overflowing* garbage in Indian streets. CityLens uses **YOLOE**, a YOLO segmentation network paired with a **CLIP-style text encoder** (MobileCLIP), instead. CLIP-type models learn from millions of image–caption pairs, so images and words end up in a shared "embedding space". YOLOE compares each region of the photo with the embeddings of text prompts and detects whatever the prompt describes.

- **Garbage prompts:** "overflowing garbage", "garbage bag", "litter".
- **Decoy prompts:** "garbage bin" and "dumpster". These soak up *empty* bins, which would otherwise be mistaken for garbage, and the backend then ignores them. Without decoys, an empty bin scored 0.59 as "pile of garbage". With them, bins score 0.84–0.86 as "dumpster" and are correctly ignored.
- The text embeddings are computed once and stored inside the model file, so the server needs no text encoder at run time.

Step 5 · Semantic segmentation: road or footpath?

A detector can say "there is a car", but not whether the car is in a lane or blocking the footpath. **SegFormer-B0**, a transformer-based segmentation model trained on the Cityscapes street-scene dataset, labels *every pixel* as road, sidewalk, building, vegetation and so on.

- **Footpath parking:** CityLens looks at the strip where each vehicle's tyres meet the ground and counts only road and sidewalk pixels there. If at least 50% of that ground contact is sidewalk, the vehicle is parked on the footpath.
- **Road-surface check:** a pothole must lie on road or pavement pixels. A "pothole" on a wall or a car roof is rejected.

Step 6 · Reading signs: OCR plus symbol detection

A car under a NO PARKING board is illegally parked, even on the road. CityLens looks for such boards in two independent ways:

Text: OCR

EasyOCR first finds text regions with the CRAFT text detector. It then reads them with a recurrent recognition network, in English and Kannada.

- Lines that sit together on one board ("NO" above "PARKING") are grouped.
- Each board is matched against phrases like NO PARKING, DO NOT PARK, TOW AWAY ZONE and ವಾಹನ ನಿಲುಗಡೆ ನಿಷೇಧ.
- Matching uses **approximate string matching** (edit distance), so "NO PARKING" still counts but plain "PARKING" never does.

Symbol: classical computer vision

The standard Indian no-parking sign has no words: a **blue disc, red ring and red diagonal slash**.

- The photo is converted to **HSV colour space**, and red and blue masks are extracted.
- Candidate circles come from the colour blobs (contours and minimum enclosing circles).
- Each candidate is scored on all three parts, and only a sign with all of them counts. On 49 photos without the sign, nothing was found.

Step 7 · Reasoning across the models

Each model only knows its own classes, so they can disagree in ways a person never would. The analysis stage settles this with geometry:

- **A pothole can't be on a car.** Seen from above, a dark car roof looks like a pothole. A pothole box that is 30% or more inside a detected vehicle is rejected.
- **Vehicles are detected at a lower confidence (0.2)** so that a car the model is less sure of can still rule out a pothole on its roof. These low-confidence vehicles are never reported as parking violations themselves.
- **Legally parked cars aren't a complaint.** When cars are found but none breaks a rule, the app explains what it saw instead of inventing a violation.

07 · DETECTION LOGIC

Detection logic, issue by issue

The AI's raw output goes through an explainable decision layer. Each rule can be tuned per city, and each decision comes with a reason the citizen and the officer can read.

Choosing the issue

Every detection whose confidence clears the threshold becomes a candidate: 0.35 for trained models, and 0.20 for the open-vocabulary garbage model, whose scores run lower even when it is right. After the cross-checks, the issue with the **most confident detection** wins. All of that issue's detections are used for severity.

When does a vehicle count as illegal parking?

A detected vehicle is reported only if **at least one** of these holds:

CONDITION	HOW IT IS MEASURED	EFFECT ON SEVERITY
Inside a notified no-parking zone	GPS distance to geofenced zones set by the city (e.g. junctions, bus stands)	Raised one level, and the zone is named in the reason
A no-parking sign is visible	OCR text or the sign symbol	Raised one level, and the sign text is quoted
Parked on the footpath	Scene segmentation: 50% or more of the ground contact is sidewalk	Raised one level
Obstructing / the subject of the photo	Vehicle covers 12% or more of the frame	Base severity

Severity engine

Severity comes from measurable evidence in the photo: **how much area the problem covers** (from the segmentation mask when available) and **how many instances** there are. Context such as the zone, a sign or the footpath raises the level further for parking.

ISSUE	AREA MEASURED	HIGH IF AREA ≥	MEDIUM IF AREA ≥	HIGH / MEDIUM IF COUNT ≥
Garbage dumping	Union of all garbage masks	20%	7%	4 / 2
Pothole	Largest pothole mask	12%	4%	3 / 2
Illegal parking	Largest vehicle (road blocked)	25%	10%	3 / 2

The result is Low Medium High, together with a plain-language reason. For example: "Vehicle occupies about 5% of the frame; 7 vehicles detected. A sign reading "NO PARKING" is visible in the photo."

Why an explainable rule layer on top of the AI? Municipal decisions must be defensible. An officer, or a citizen whose complaint was ranked "Low", can see exactly why. The thresholds can be adjusted for each city without retraining any model, and the same evidence always gives the same rating.

The citizen in the loop

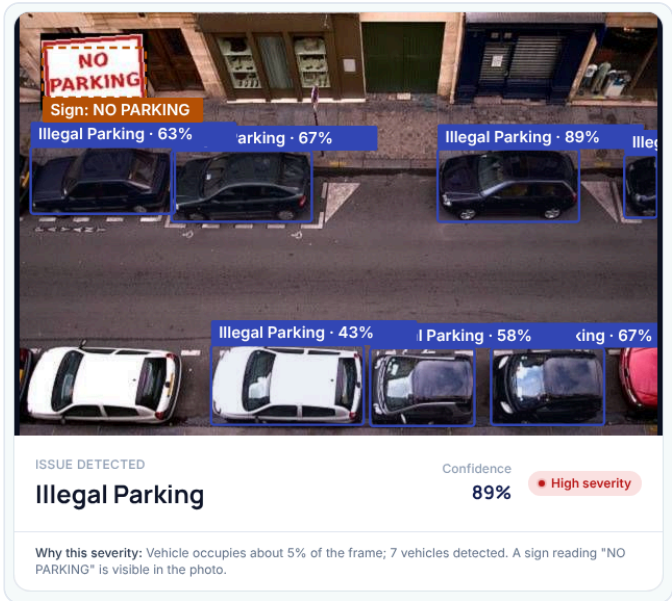
The citizen confirms or corrects every AI result before a complaint is filed, and can choose the issue manually when the AI finds nothing. Each complaint records whether it was *AI-detected* or *corrected by the citizen*, which gives the city an accuracy measure and a labelled dataset for improving the models.

Results on real photos

Examples of the pipeline running end to end on real street photos.



Garbage dumping: High severity. The open-vocabulary model outlines the overflowing bin and the bags spilling onto the footpath (green): 6 masks covering about 24% of the photo, confidence 62%. The empty bin on the left is matched to the "dumpster" decoy and ignored.



Illegal parking: High severity. The cars here are on the road, small in the frame and outside any zone, but OCR reads the NO PARKING board, so they are reported. In the same photo without the board, no violation is reported. In this test the board was added to the photo.

TEST PHOTO	MODELS INVOLVED	RESULT
Overflowing bin next to an empty bin	YOLOE garbage	Garbage dumping, High, 62%. Empty bin ignored
Top-down street, cars in marked paid bays	YOLOv8s + TTA, pothole, cross-checks	No violation. Explains that 9 legally parked vehicles were found. The earlier false "pothole" on a car roof is now rejected
Same street with a NO PARKING board	+ EasyOCR	Illegal parking, High: the sign reading "NO PARKING" is quoted
Same street with the no-parking symbol	+ OpenCV symbol detector	Illegal parking, High: no-parking sign visible
Same street with a PARKING board	+ EasyOCR	No violation. Explanation mentions the parking sign
Pothole in an open road	Pothole model	Pothole detected and routed to Roads

Built for public-sector trust

Identity and access

- Every request carries a signed JWT. A user's identity is never read from the request itself.
- Citizens see only their own complaints.
- Officers see and act only on their own department's queue.
- Officer accounts can't be self-registered. Public sign-up always creates a citizen account.
- Passwords are stored as PBKDF2-SHA256 hashes with a salt for each user.

Workflow integrity

- Status moves only forward. Resolving or rejecting a complaint requires a note.
- Closed complaints are locked.
- The officer's name on each action comes from their sign-in, so actions can't be signed in someone else's name.
- "After" photos provide visual proof that the fix was done.
- The server decides the routing, so a complaint can't be misrouted.

Testing. 82 automated tests check these rules directly. They cover cross-citizen and cross-department access attempts, forged and expired tokens, spoofed officer names and the detection cross-checks.

Privacy by design

- Only the photo, its location and the description are collected. Citizens' contact details are never shown to officers beyond what the complaint needs.
- Planned for production: automatic blurring of faces and number plates in stored photos, data retention limits, and hosting in India in line with the Digital Personal Data Protection Act, 2023.

10 · BUSINESS MODEL

Who pays, and why

The model is B2G SaaS: the city pays, and citizens use it for free. Every complaint that arrives verified, located and rated saves field-inspection time and speeds up the fix.

SEGMENT	WHO	VALUE TO THEM
Primary customer	Municipal corporations, city municipal councils and Smart City SPVs	Verified, ranked complaints; fewer inspection trips; data on response times for accountability
Secondary customers	Traffic police, development authorities, highway and road maintenance contractors	Parking enforcement evidence; road-condition monitoring
Private sites	Tech parks, university campuses, gated communities, industrial estates	The same loop for their facility-management teams
Free users	Citizens	Report in a minute, see the problem fixed

Business model canvas

Value proposition

AI-verified, geotagged, severity-ranked civic complaints routed to the right department, with proof of resolution.

Revenue streams

- Monthly or annual SaaS subscription by city size
- One-time setup and integration
- On-premise deployment and support
- Analytics reports and custom model tuning

Channels

- Free 90-day pilot in one ward
- Government e-Marketplace (GeM) listing
- Smart City Mission and state urban development departments
- Hackathons and civic-tech programmes

Key activities

- Model improvement on local photos
- Onboarding and training officers
- Configuring zones, departments and languages per city

Key resources

- The vision pipeline and decision layer
- Citizen-confirmed photo dataset, which grows with use
- The engineering team

Cost structure

- Cloud computing (CPU/GPU) and storage
- Support and onboarding staff
- Development
- No model licence fees (open source)

Market size: a bottom-up estimate

India has **279 municipal corporations**, 19 of them in Karnataka, including Mangaluru City Corporation with 60 wards and a population of 7.24 lakh (2011 Census) ([list of municipal corporations](#)). Thousands of smaller municipal councils and town panchayats come on top of that.

SCENARIO	CORPORATIONS	PRICE / MONTH	ANNUAL REVENUE
Beachhead: 5 Karnataka corporations	5	₹79,999	≈ ₹48 lakh
10% of India's corporations	28	₹79,999	≈ ₹2.7 crore
25% of corporations	70	₹79,999	≈ ₹6.7 crore

These are illustrative estimates for corporations only, before smaller councils, private sites and add-ons.

11 · PRICING FOR AUTHORITIES

Subscription plans

Prices are monthly and exclude GST. Plans are sized by city population, so a small council pays a fraction of what a metro pays.

Pilot Any city, one ward, 90 days Free • 1 department • 5 officer seats • Full AI pipeline • Success report at the end	Town Town panchayats and municipalities under 1 lakh people ₹14,999 /mo • 3 departments • 10 officer seats • 3,000 AI analyses / month • Email support	City City municipal councils, 1–5 lakh people ₹39,999 /mo • All departments • 25 officer seats • 15,000 analyses / month • Response-time (SLA) dashboards • Kannada / Hindi OCR	Corporation Municipal corporations, 5–15 lakh (e.g. Mangaluru) ₹79,999 /mo • Unlimited seats • 50,000 analyses / month • Ward-level analytics • Custom zones and model tuning • Priority support
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PLAN / ADD-ON	PRICE	NOTES
Metro / Enterprise (over 15 lakh people)	From ₹2,00,000 / month	Custom limits, on-premise GPU option, dedicated support
Annual billing	Pay for 10 months, get 12	About a 17% discount; suits government budget cycles
Setup and onboarding	₹50,000 one-time (waived after a pilot)	Department setup, no-parking zones, officer training
Integration with existing grievance systems	₹1.5 lakh one-time	Push complaints into the city's current portal or helpline
On-premise deployment	₹3 lakh one-time + 20% yearly support	For cities that require data to stay on their own servers
Private sites (campuses, tech parks)	₹9,999 / month	Up to 5 seats, one site

Why these prices are justified

- **Benchmark.** Vialytics, an AI road-assessment service used by small US towns, is quoted at **\$12,500 a year with a three-year commitment** ([Dubois County Free Press](#)). That covers road condition only. The CityLens Corporation plan costs ₹9.6 lakh a year and covers three issue types plus the complaint workflow.
- **Small next to a city budget.** For a corporation of about 7 lakh people, ₹9.6 lakh a year is roughly ₹1.3 per resident per year.

Unit economics (estimates)

ITEM, PER CORPORATION CUSTOMER PER MONTH	ESTIMATE
Revenue	₹79,999
Cloud computing: one 8-vCPU server handles about 3–4 s of analysis per photo, enough for 50,000 photos a month	₹10,000 – 15,000
Storage, bandwidth and backups, plus a share of one support person	₹13,000 – 20,000
Gross margin	≈ 55–70%

SWOT analysis

Strengths

- Combines several vision techniques: detection, two kinds of segmentation, open-vocabulary detection and OCR
- Understands context: footpath, zones, signs and cross-model checks
- Every severity rating comes with a written reason
- Works from one photo on any phone; no special hardware
- Open-source stack with no licence fees
- Closed loop with proof of resolution
- Supports Kannada as well as English

Weaknesses

- Pretrained models aren't yet tuned on local Mangaluru photos
- Accuracy drops in night, rain and blurry photos
- Only three issue types for now
- Depends on citizens to take part
- OCR treats time-limited boards ("8 AM–8 PM") as always applying
- Student team without a public-sector sales track record

Opportunities

- Smart City and digital-governance push across India
- 279 municipal corporations plus thousands of smaller councils
- Citizen-confirmed photos become a unique local dataset
- New issue types: streetlights, waterlogging, stray animals, fallen trees
- Dashcam and bus-camera feeds for passive road surveys
- Ward-level analytics to plan budgets

Threats

- Slow government procurement and budget cycles
- Large IT vendors bundling grievance modules
- Free state-run apps that citizens already know
- Privacy rules (DPDP Act) around faces and number plates
- Fake or spam reports
- Officer resistance if dashboards expose slow responses

How we address the main risks

RISK	RESPONSE
Accuracy on local photos	Pilot data with citizen corrections → fine-tune on local images → measure precision and recall before scaling
Slow procurement	Free pilot, GeM listing, low-cost Town plan that fits within smaller discretionary budgets
Spam / fake reports	Sign-in required; GPS from the photo; AI verification; duplicate detection by location (roadmap)
Privacy	Blur faces and number plates, host in India, data retention limits

What comes next

PHASE	MILESTONES
Now	Working citizen app and authority portal; six-model vision pipeline; authentication; 82 tests
Next 3 months	One-ward pilot; collect local photos; fine-tune the pothole and garbage models; measure accuracy against citizen confirmations
3–6 months	Duplicate merging by location; face and number-plate blurring; Android app; WhatsApp reporting
6–12 months	More issue types; ward analytics; integration with city grievance portals; GeM listing

Go-to-market path. Free pilot in one Mangaluru ward → measured results (complaints, response time, AI accuracy confirmed by citizens) → Corporation plan → reference case for the other 18 Karnataka corporations → GeM listing for the rest of India.

Likely questions, with answers

The questions judges, faculty and city officials are most likely to ask.

TECHNOLOGY AND COMPUTER VISION

Q1. Why YOLO instead of other detectors like Faster R-CNN?

YOLO is a single-stage detector: one network pass per image. That makes it fast enough to answer while the citizen waits, even without a GPU. Two-stage detectors such as Faster R-CNN can be slightly more accurate on tiny objects but are several times slower. YOLOv8 also gives detection and segmentation in one family.

Q2. What is the difference between object detection, instance segmentation and semantic segmentation in your project?

Detection draws a box around each object: "a car here". Instance segmentation outlines each object pixel by pixel: "this pothole's exact shape", which we use to measure real area for severity. Semantic segmentation labels every pixel by category: "this is road, that is footpath", which tells us where a vehicle is standing.

Q3. How do you detect garbage without a garbage dataset?

With open-vocabulary detection. YOLOE pairs a YOLO network with a CLIP text encoder, so we describe garbage in words ("overflowing garbage", "garbage bag", "litter") instead of labelling thousands of images. "Decoy" prompts for bins stop empty bins being reported. As citizens confirm reports, we will collect a labelled local dataset and fine-tune.

Q4. How do you decide if parking is illegal? A car on the road isn't always illegal.

Exactly, so we don't flag every car. A vehicle is reported only if it is on the footpath (semantic segmentation), inside a no-parking zone (GPS geofence), under a NO PARKING sign (OCR or sign-symbol detection), or large in the frame, meaning it is the obstruction being reported. Otherwise the app says the cars look legally parked.

Q5. What happens when the AI is wrong?

Nothing is filed until the citizen confirms. They can correct the issue type, and the complaint records that it was corrected. Within the pipeline, the models also check each other. For example, a "pothole" on top of a detected car is rejected. Corrections are logged, which gives us an accuracy measure and training data.

Q6. How is severity calculated? Isn't it subjective?

It is measured, not guessed: the area the problem covers in the photo (from the segmentation mask) and how many instances there are. For parking, being on the footpath, in a zone or under a sign raises it. Every rating comes with a written reason, and cities can tune the thresholds.

Q7. What is your model accuracy?

We use pretrained public models, each with published benchmark results. On our own test photos, the end-to-end decisions were correct in every case shown in this report. The honest answer for a city is that we will measure precision and recall during the pilot, using citizen confirmations as ground truth, and fine-tune on local photos.

Q8. Does it work at night or in the rain?

Accuracy is lower in poor light. CityLens has an optional contrast-enhancement step (CLAHE), and the citizen confirmation step catches mistakes. Collecting local night-time photos for fine-tuning is part of the pilot plan.

Q9. How fast is it, and does it need a GPU?

It runs on an ordinary CPU server. A photo takes a few seconds, including sign reading when cars are present, and much less on a GPU. The scene segmentation and OCR steps only run when a vehicle is detected, which keeps the average time low.

Q10. How do you get the location?

From the GPS data stored inside the photo (EXIF), or from the device's location when the photo has none. The coordinates are turned into an address with OpenStreetMap, and complaints appear on the authority map.

PRODUCT, USERS AND SECURITY

Q11. How do officers get their accounts? Can anyone sign up as an officer?

No. Public sign-up always creates a citizen account. The city administrator creates officer accounts for each department. Each officer sees only their own department's complaints, and every action is recorded under their name from their sign-in.

Q12. How do you stop fake or spam complaints?

Users must sign in. The location comes from the photo itself, and the AI must find a supported problem, otherwise the citizen has to choose it manually, which is flagged. Merging duplicates by location is on the roadmap.

Q13. What about privacy? Photos contain faces and number plates.

We collect only what a complaint needs. For production, we plan automatic blurring of faces and plates, India-hosted storage, and retention limits in line with the DPDP Act, 2023.

Q14. Why would citizens use this instead of calling the helpline?

It's faster: one photo, no typing addresses or choosing departments. They also see progress, with a status timeline and an after-fix photo. Visible results are what make people report again.

Q15. How is this different from the Swachhata app or existing city portals?

Those apps collect photos, but people sort and review them, and each covers one department. CityLens recognises the issue automatically, measures severity, understands context such as footpaths and signs, and routes across departments.

BUSINESS

Q16. Who pays for this?

The city: municipal corporations, councils, Smart City SPVs and traffic departments, through a monthly or annual subscription. Citizens use it for free. Private campuses and tech parks are a second, faster-moving market.

Q17. How much does it cost a city?

From ₹14,999 a month for small towns to ₹79,999 a month for a municipal corporation like Mangaluru, about ₹1.3 per resident per year. There is a free 90-day pilot in one ward first.

Q18. Government sales are slow. How will you get your first customer?

With a free pilot in one ward, which needs no procurement, and success metrics agreed up front. A paid plan follows from those results, and the reference case is used across Karnataka's 19 corporations. GeM listing makes later purchases simple.

Q19. What is your competitive advantage? Can't a big company copy this?

Our advantage is the combination: an understanding of the whole scene, explainable severity, Indian context such as Kannada OCR and Indian sign symbols, and the closed loop. Over time, the strongest advantage is the dataset of citizen-confirmed local photos, which grows with every city.

Q20. What does it cost you to run?

We estimate ₹25,000–35,000 a month per corporation customer, covering cloud computing, storage and support. That gives a gross margin of about 55–70%. There are no model licence fees because the stack is open source.

Q21. How will you measure impact?

By the number of complaints, the average time to resolution per department, the share resolved with photo proof, the AI accuracy confirmed by citizens, and repeat reporting by the same citizens.

Q22. What would you build with funding?

A pilot with a real city, fine-tuning on local photos, a native Android app and WhatsApp reporting, blurring of faces and plates, and duplicate merging.

A 30-second pitch. "Every day people walk past potholes, garbage and cars blocking footpaths, and almost nobody reports them, because reporting is slow and nothing happens. CityLens turns one photo into a verified complaint. Computer vision identifies the problem, measures how serious it is, understands context like footpaths and no-parking signs, and sends it straight to the right department. The citizen tracks it until there's a photo of the fix. It's free for citizens; cities pay from ₹15,000 a month, and the first ward pilot is free."

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Pricing, market-size and cost figures in sections 10–11 are the team's proposals and estimates for planning and pitching. They are not quotes from a signed customer.